

Example

SI No	Age	Income	Student	Credit rating	Buy Comp C;
1	(35)	(Medium)	Yes	Fair	(Yes)
2	30	High	No	Average	No
3	40	Low	Yes	Good	No
4	(35)	(Medium)	No	Fair	(Yes)
5	45	Low	No	Fair	(Yes)
6	35	High	No	Excellent	(Yes)
7	(35)	(Medium)	No	Good	No *
8	25	Low	No	Good	No
9	28	High	No	Average	No
10	(35)	(Medium)	Yes	Average	(Yes)

→ Determine whether a ~~student~~ customer with 35 years old and medium income will buy a computer or not. Use Bayesian classifier.

Ans:

$$X_1 = \text{Age}, X_2 = \text{Income} \quad 35 = 35 \text{ (A)} \quad 9 = (9|X)9$$

$$X = X_1 \cap X_2 = (\text{Age} = 35) \cap (\text{Income} = \text{medium})$$

$$C_1 = \text{will buy a computer}$$

$$C_2 = \text{Will not buy a computer}$$

From Bayesian equation,

$$\text{if } A = C_1 \\ B = X$$

$$P(C_1|X) = \frac{P(X|C_1) \cdot P(C_1)}{P(X)} \dots \dots \dots \text{Eq ①}$$

$$\text{if } A = C_2 \\ B = X$$

$$P(C_2|X) = \frac{P(X|C_2) \cdot P(C_2)}{P(X)} \dots \dots \dots \text{Eq ②}$$

$$\text{For Eq ①} \Rightarrow P(C_1) = P(\text{will buy a computer}) = 5/10$$

$$P(X) = P(X_1 \cap X_2)$$

$$= P(\text{Age} = 35 \cap \text{Income} = \text{medium}) \\ = 4/10$$

$$P(X|C_1) = P(\text{Age} = 35 \wedge \text{Income} = \text{medium} | \text{will buy a computer})$$

$$= \frac{3}{5}$$

$$\therefore P(C_1|X) = \frac{P(X|C_1) P(C_1)}{P(X)}$$

$$= \frac{(3/5) * (5/10)}{4/10}$$

$$= 3/4 = 0.75$$

for eq 2 $\Rightarrow$

$$P(C_2) = P(\text{will not buy a computer}) = 5/10$$

$$P(X) = P(\text{Age} = 35 \wedge \text{Income} = \text{medium}) = 4/10$$

$$P(X|C_2) = P(\text{Age} = 35 \wedge \text{Income} = \text{medium} | \text{will not buy a computer})$$

$$= 1/5$$

$$\begin{aligned}
 \therefore P(C_2|x) &= \frac{P(x|C_2) P(C_2)}{P(x)} \\
 &= \frac{\frac{1}{5} \times \frac{8}{10}}{\frac{4}{10}} = \frac{1}{4} = 0.25
 \end{aligned}$$

$$\begin{aligned}
 \text{Final outcome} &= \operatorname{argmax} (P(C_1|x), P(C_2|x)) \\
 &= \operatorname{argmax} (0.75, 0.25)
 \end{aligned}$$

$(C_1) \Rightarrow (C_1) = C_1$ will buy a computer.

Naive Bayesian Classification

$$(1) P(x, y|z) = \frac{P(x, y, z)}{P(z)}$$

$$(2) P(x, y|z) = \frac{P(x, y, z)}{P(y, z)} * \frac{P(y, z)}{P(z)}$$

$$(3) P(x, y|z) = P(x|y, z) * P(y|z)$$

$(4) P(x, y|z) = P(x|z) * P(y|z)$ when x and y are conditionally independent

✓ Example

Solve the previous example using the Naive Bayes Classifier.

Ans

$x_1 = \text{"Age"} = 35$, $x_2 = \text{"Income"} = \text{medium}$

$C_1 = \text{"will buy a computer"}$, $C_2 = \text{"will not buy a computer"}$

$x = x_1 \wedge x_2$

$$P(C_1|x) = \frac{P(x|C_1) P(C_1)}{P(x)} = \frac{P(x_1, x_2|C_1) P(C_1)}{P(x)}$$

$$= \frac{P(x_1|C_1) P(x_2|C_1) P(C_1)}{P(x)}$$

$$= (0.5 \times 0.4 \times 0.6) \text{ eq (1)}$$

Again,

$$P(C_2|x) = \frac{P(x|C_2) P(C_2)}{P(x)} = \frac{P(x_1, x_2|C_2) P(C_2)}{P(x)}$$

$$= \frac{P(x_1|C_2) P(x_2|C_2) P(C_2)}{P(x)}$$

$$\dots \dots \dots \text{eq (2)}$$

for eq (1) $\Rightarrow P(C_1) = P(\text{will buy a computer}) = 5/10$

$$P(x_1|C_1) = P(\text{Age} = 35 | \text{will buy a computer}) = 4/5$$

$$P(x_2|C_1) = P(\text{Income} = \text{medium} | \text{will buy a comp}) = 3/5$$

for eq (2) $\Rightarrow P(C_2) = P(\text{will not buy a computer}) = 5/10$

$$P(x_1|C_2) = P(\text{Age} = 35 | \text{will not buy a computer}) = 1/5$$

$$P(x_2|C_2) = P(\text{Income} = \text{medium} | \text{will not buy a computer}) = 1/5$$

$$\begin{aligned} \text{Final output class} &= \arg\max (P(c_1|x), P(c_2|x)) \\ &= \arg\max \left(\frac{P(x_1|c_1) P(x_2|c_1) P(c_1)}{P(x)}, \frac{P(x_1|c_2) P(x_2|c_2) P(c_2)}{P(x)} \right) \end{aligned}$$

$$\begin{aligned} &\cong \arg\max (P(x_1|c_1) P(x_2|c_1) P(c_1), P(x_1|c_2) P(x_2|c_2) P(c_2)) \\ &= \arg\max \left(\frac{4}{5} * \frac{3}{5} * \frac{5}{10}, \frac{1}{5} * \frac{1}{5} * \frac{5}{10} \right) \end{aligned}$$

$$= c_1$$

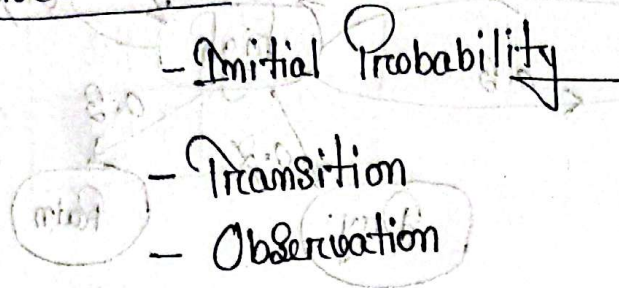
= Will buy a computer

Ans

Date: 10-10-25

Hidden Markov Model

→ Model Needs:



Mid Report Submission

23/10/25 (Thursday) 7:30

24/10/25 Report Submission

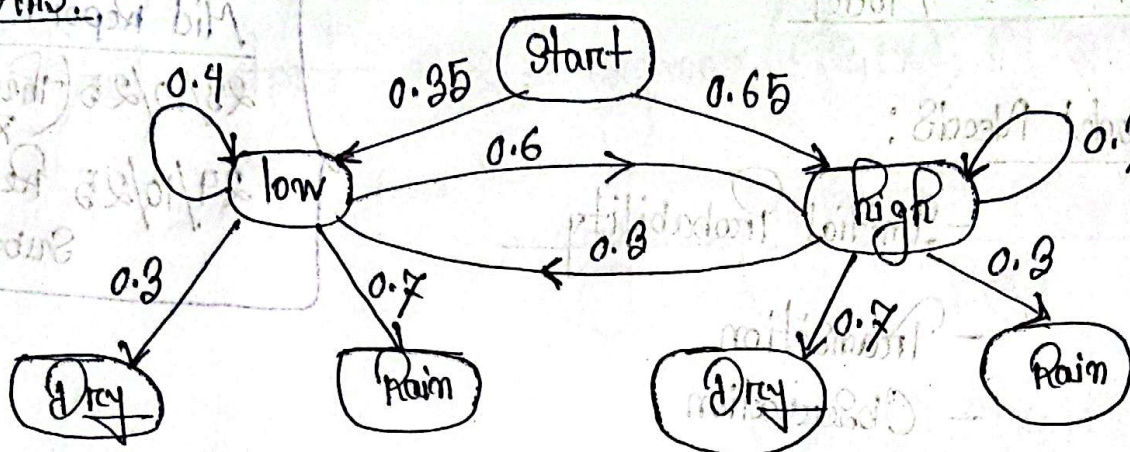
Ques

For forecasting weather, it is observed that low atmospheric pressure causes a chance of 70% Rain, otherwise Dry. On the other hand, 70% chance of Dryness is happened if high atmospheric pressure prevails. Again there is a 60% probability of changing from low to high atmospheric pressure, but the high to low changing occurred in a probability of 30%. the initial probability of high atmospheric pressure is 65%.

① Draw a Hidden Markov Model

② If a rainy weather prevails on today, the cricket match on tomorrow will be happen or not.

Ans:



Initial Probability:

$$P(\text{low} | \text{Start}) = 0.35$$

$$P(\text{high} | \text{Start}) = 0.65$$

Transition Probability

$$P(\text{low} | \text{low}) = 0.4$$

$$P(\text{high} | \text{low}) = 0.6$$

$$P(\text{high} | \text{high}) = 0.7$$

$$P(\text{low} | \text{high}) = 0.3$$

Transition Matrix:

	low	high
low	0.4	0.6
high	0.3	0.7

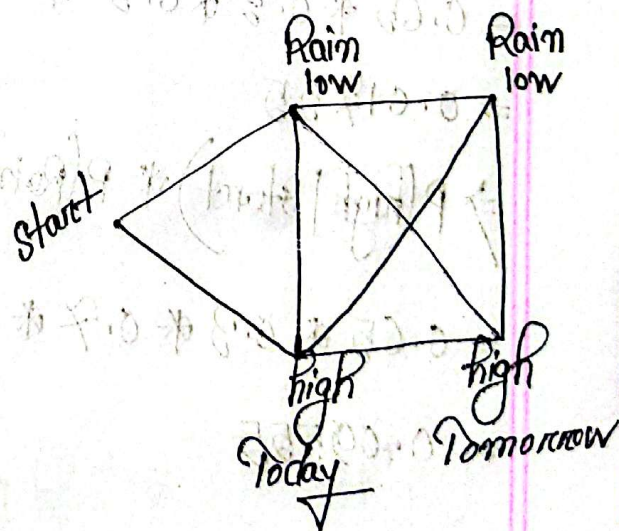
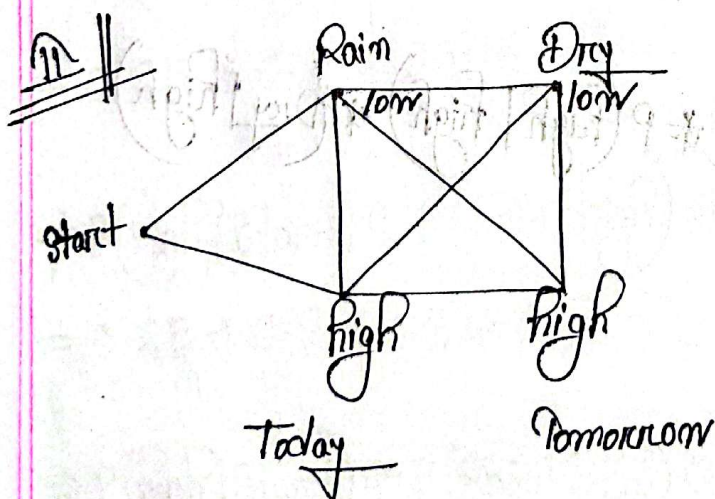
Observation Probability

$$P(\text{Dry} | \text{low}) = 0.3$$

$$P(\text{Rain} | \text{low}) = 0.7$$

$$P(\text{Dry} | \text{high}) = 0.7$$

$$P(\text{Rain} | \text{high}) = 0.3$$



Hidden State Sequence

Start $\rightarrow$ low $\rightarrow$ low

Start $\rightarrow$ low $\rightarrow$ high

Start $\rightarrow$ high $\rightarrow$ low

Start $\rightarrow$ high $\rightarrow$ high

Path Probabilities

$$\Rightarrow P(\text{low} | \text{Start}) * P(\text{Rain} | \text{low}) * P(\text{low} | \text{low}) * P(\text{Dry} | \text{low})$$

$$= 0.35 * 0.7 * 0.4 * 0.3$$

=

$$\Rightarrow P(\text{low} | \text{Start}) * P(\text{Rain} | \text{low}) * P(\text{high} | \text{low}) * P(\text{Dry} | \text{high})$$

$$= 0.35 * 0.7 * 0.6 * 0.7$$

$$= 0.1029$$

$$\Rightarrow P(\text{high} | \text{Start}) * P(\text{Rain} | \text{high}) * P(\text{low} | \text{high}) * P(\text{Dry} | \text{low})$$

$$= 0.65 * 0.3 * 0.3 * 0.3$$

$$= 0.01755$$

$$\Rightarrow P(\text{high} | \text{Start}) * P(\text{Rain} | \text{high}) * P(\text{high} | \text{high}) * P(\text{Dry} | \text{high})$$

$$= 0.65 * 0.3 * 0.7 * 0.7$$

$$= 0.09555$$

$$\therefore \text{Total Probability} = 0.2454$$

Similarly $\rightarrow$ (2nd graph कि छन नैन सनर)

Hidden State Sequence

Start $\rightarrow$ low $\rightarrow$ low

Start $\rightarrow$ low $\rightarrow$ high

Start $\rightarrow$ high $\rightarrow$ low

Start $\rightarrow$ high $\rightarrow$ high

After calculate path probability
Total Probability = 0.1946

Since $P(\text{Rain Dry}) > P(\text{Rain Rain})$
 $\therefore$ Tomorrow Dry Cricket match
will happen

$$\Rightarrow P(\text{low} | \text{Start}) * P(\text{Rain} | \text{Low}) * P(\text{Low} | \text{Low}) * P(\text{Rain} | \text{Low})$$
$$= 0.35 * 0.7 * 0.4 * 0.7$$

$$\Rightarrow P(\text{low} | \text{Start}) * P(\text{Rain} | \text{low}) * P(\text{High} | \text{low}) * P(\text{Rain} | \text{high})$$
$$= 0.35 * 0.7 * 0.6 * 0.3$$

$$\Rightarrow P(\text{high} | \text{Start}) * P(\text{Rain} | \text{high}) * P(\text{low} | \text{high}) * P(\text{Rain} | \text{low})$$
$$= 0.65 * 0.3 * 0.3 * 0.7$$

$$\Rightarrow P(\text{high} | \text{Start}) * P(\text{Rain} | \text{high}) * P(\text{high} | \text{high}) * P(\text{Rain} | \text{high})$$
$$= 0.65 * 0.3 * 0.7 * 0.3$$